

**23CSE452 - Business Analytics**  
**Individual Case Study Report**

## **Influencer Marketing Analytics: Predicting High-Engagement YouTube Campaigns for Marketing Investment Optimization**

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**Business Domain: Social Media Marketing**

# 1. Problem Statement and Objectives

## 1.1 Problem Statement

Businesses are putting a growing share of their marketing budgets into YouTube influencer campaigns, but it is still difficult to tell before investing whether a campaign will succeed. In practice, most organizations choose influencers based on subscriber count or general popularity. They pay much less attention to the content itself and to how the creator's audience has responded to earlier videos, even though these are more likely to show how effective a campaign will be.

In this study I analyse YouTube video and channel data that I collected myself. The aim is to find out which factors influence audience engagement, and to build a model that can estimate whether a video is likely to achieve high engagement before the investment decision is made.

## 1.2 Importance of the Problem

Influencer fees are usually negotiated on reach, which in practice means the size of the creator's audience. If reach is only weakly related to how many viewers actually respond to the content, then brands are paying for the wrong thing. A sponsored video that fails to engage its audience wastes the whole fee, so even a small improvement in how creators are shortlisted can improve the return on marketing spend without increasing the budget. There is also a practical constraint: the brand has to decide before the sponsored video exists, so any prediction can only use information that is available at that point. This requirement shaped most of the decisions in this study.

## 1.3 Objectives

1. To identify the channel, content and timing factors that influence audience engagement on YouTube, using only information that is available before investing.
2. To develop a classification model that predicts whether a video is likely to achieve high engagement, and to compare it with the common practice of selecting influencers by subscriber count.
3. To use the results to support influencer selection and marketing budget allocation, and to give practical recommendations for campaign planning.

# 2. Data Collection and Dataset Description

## 2.1 Data Source and Collection Method

The data was collected from publicly available YouTube channels. My proposal planned to scrape the data using Apify, and I started with the Apify actor `apidojo/youtube-scraper` [8]. However, every test run returned exactly 10 records, even when I requested 250. The run log showed that the actor limits Apify Free Plan accounts to 10 items per run. At that rate, collecting 10,000 records would have needed about 1,000 separate runs, so I switched to the official YouTube Data API v3 provided by Google [6]. The API returns the same public information that is shown on YouTube's pages, and it has one advantage over the scraper: like and comment counts are exact, whereas the scraper rounded comment counts (for example to 4,600). The files from the Apify attempt are kept in the project repository as a record.

I wrote Python scripts to collect the data through the API. For each channel, the script first calls `channels.list` to get the subscriber count, the number of uploads and the ID of the channel's uploads playlist. It then calls `playlistItems.list` to get the IDs of the 150 most recent uploads, and finally `videos.list` to download the statistics and details of those videos in batches of 50. The data was saved after each channel, so a run could be restarted if it stopped part way. Two more passes were made later: one to record the declared audio language of every video, and one to collect older long-form videos for 21 channels whose recent uploads were almost all YouTube Shorts.

Collection used 3,091 API quota units in total, well within the free limit of 10,000 units per day, so the data cost nothing to collect. Table 1 summarises the collection runs.

**Table 1:** Summary of data collection

| Collection step                                                  | Records       | Cost              |
|------------------------------------------------------------------|---------------|-------------------|
| Apify actor, 3 test runs (abandoned because of the 10-row limit) | 30            | \$0.015           |
| YouTube Data API: up to 150 recent uploads per channel           | 22,442        | Free              |
| YouTube Data API: older long-form videos for 21 channels         | 365           | Free              |
| <b>Total records used</b>                                        | <b>22,807</b> | 3,091 quota units |

## 2.2 Selection of Channels

I first selected 100 well-known channels by hand, 10 from each of 10 niches: tech, gaming, beauty, fitness, food, finance, travel, education, fashion and lifestyle. Ten of these had to be removed because the handle could not be found or pointed to a different channel. When I checked the remaining channels, 85 of the 91 that resolved had more than one million subscribers. This was a problem, because the main question of the study is whether subscriber count predicts engagement, and that cannot be tested properly if almost every channel is large. To fix this, I used the API's search function to find 60 more channels: three per niche with fewer than 200,000 subscribers and three per niche with 200,000 to one million. The final sampling frame had 150 channels covering all 10 niches and four subscriber tiers.

## 2.3 Dataset Size and Attributes

In total, 22,807 video records were collected. After cleaning (Section 3.1), 11,877 long-form videos from 143 channels were used in the analysis. Both numbers are above the minimum of 10,000 records required for this case study. Each record has 20 attributes, and Table 2 lists the ones that matter most for this study.

**Table 2:** Important variables in the dataset

| Variable                     | Description                      | Use in this study                                           |
|------------------------------|----------------------------------|-------------------------------------------------------------|
| views, likes, comments       | Audience response counts         | Used only to calculate the target variable                  |
| subscriberCount              | Subscribers of the channel       | Main variable tested against the industry practice          |
| channelVideoCount            | Total uploads of the channel     | Shows how established the channel is                        |
| duration                     | Video length in seconds          | Separates Shorts from long-form videos; used as a predictor |
| publishDate                  | Publication date and time (UTC)  | Gives upload day, time of day and video age                 |
| title, description, keywords | Text written by the creator      | Used to create title and description features               |
| niche                        | Niche of the channel             | Predictor and grouping variable                             |
| defaultAudioLanguage         | Language declared by the creator | Used to remove non-English channels                         |

The other eight attributes are id, channelName, channelHandle, channelId, url and type, which identify each record, and category and madeForKids, which were examined and then dropped for the reasons given in Section 3.2.

All of the data is public channel and video information, and no viewer data or comment text was collected. Some creators list a business email address or phone number in their descriptions, so before submission I masked these with asterisks using a script (scripts/anonymize\_raw.py). This affected 3,979 videos and did not change any of the variables used in the analysis.

### 3. Data Preparation and Exploratory Analysis

#### 3.1 Data Cleaning and Preprocessing

All cleaning and preprocessing was done in the Jupyter notebook (`analysis.ipynb`). The notebook does not change the raw files; each step is applied in memory and the number of rows removed is recorded. Table 3 summarises the steps.

**Table 3:** Data cleaning steps

| Step                                      | Removed | Remaining | Reason                                          |
|-------------------------------------------|---------|-----------|-------------------------------------------------|
| Keep channels in sampling frame           | 150     | 22,657    | Handle resolved to the wrong channel (@minale)  |
| Drop duplicate video ids                  | 0       | 22,657    | Same video collected twice                      |
| Drop non-English channels                 | 600     | 22,057    | Declared audio language mostly non-English      |
| Drop missing views/likes/comments         | 386     | 21,671    | Likes hidden or comments disabled               |
| Drop videos under 1,000 views             | 265     | 21,406    | Engagement rate too noisy on small denominators |
| Drop videos under 30 days old             | 2,056   | 19,350    | Still accumulating views                        |
| Drop Shorts ( $\leq 180$ s)               | 6,753   | 12,597    | Different format; views counted differently     |
| Drop impossible engagement rate ( $> 1$ ) | 0       | 12,597    | More likes + comments than views - data error   |
| Drop videos with $< 5$ earlier videos     | 720     | 11,877    | No engagement history for the channel yet       |

A few of these steps need some explanation. First, I decided to analyse only long-form videos and removed YouTube Shorts (videos of 180 seconds or less). Brands usually place sponsored content inside long-form videos, and since March 2025 YouTube counts every replay of a Short as a new view, so the engagement rate of a Short is not measured in the same way as that of a normal video.

Second, four channels were removed because most of their videos were in Malayalam, Hindi or Telugu. I had asked the API for English results, but this setting only affects the ranking of search results and does not filter them. Third, videos with fewer than 1,000 views were removed because the engagement rate becomes unreliable when views are very low, and videos less than 30 days old were removed because their view counts were still growing. Missing descriptions and tags were not treated as errors, since they simply mean the creator left the field empty. They were kept as empty text and zero tags.

#### 3.2 Target Variable and Feature Engineering

The target variable is based on the engagement rate, which I defined as  $(\text{likes} + \text{comments}) / \text{views}$ . A video is labelled as *high engagement* if its engagement rate is in the top 30% of the cleaned data, which corresponds to an engagement rate of 4.52% or more. This gives 30% high-engagement videos and 70% normal videos.

From the raw attributes I created 20 predictor variables in five groups: channel variables (subscribers and total uploads, both on a log scale, the niche, and the channel's past engagement); content variables (duration on a log scale and number of tags); title variables (length, and whether the title contains a number, a question mark, an exclamation mark, capitalised words or emoji); description variables (length, number of links, hashtags, and whether a sponsor or discount code is mentioned); and timing variables (day of the week, time of day in UTC, weekend upload and the age of the video).

I was careful to avoid data leakage, which happens when the predictors contain information that would not be available at decision time or that directly reveals the answer. Views, likes and comments were not used as predictors, because the target is calculated from them. The channel's past engagement was calculated only from videos that the channel published *before* the video being predicted, with at least five earlier videos required. I did not use a simple average over all of the channel's videos, since that would include the video itself as well as videos published later.

I also removed two predictors that repeated information already in the data. Title word count had a correlation of 0.92 with title length, and the YouTube category mostly repeats the niche (97% of

gaming videos are in the Gaming category). Keeping both would have made the logistic regression coefficients unstable.

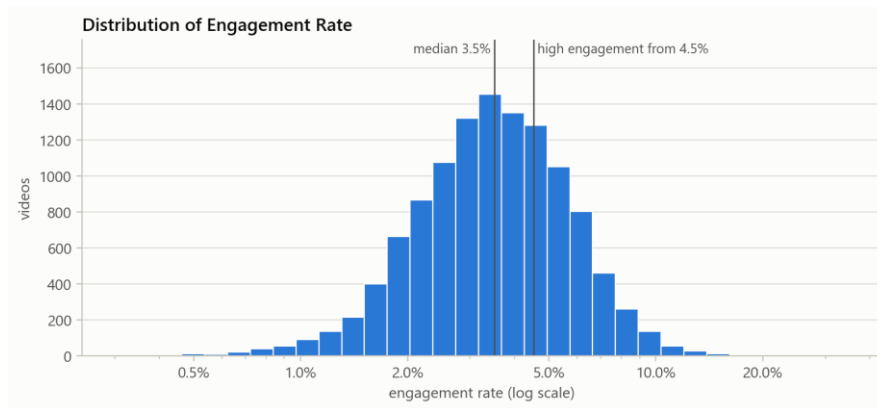
### 3.3 Descriptive Analysis

Table 4 gives the descriptive statistics of the main numeric variables after cleaning.

**Table 4:** Descriptive statistics of the cleaned dataset

| Variable                        | Mean      | Median    | Std. dev. | Min    | Max         |
|---------------------------------|-----------|-----------|-----------|--------|-------------|
| Engagement rate                 | 0.039     | 0.035     | 0.020     | 0.000  | 0.392       |
| Views                           | 1,322,074 | 229,894   | 4,180,634 | 1,004  | 161,635,321 |
| Likes                           | 35,446    | 7,972     | 78,208    | 0      | 1,524,355   |
| Comments                        | 1,881     | 452       | 4,542     | 0      | 112,968     |
| Subscribers                     | 6,232,448 | 2,620,000 | 8,733,467 | 10,000 | 45,900,000  |
| Duration (seconds)              | 1,741     | 1,074     | 2,946     | 181    | 82,302      |
| Title length (characters)       | 53        | 50        | 21        | 3      | 100         |
| Description length (characters) | 1,611     | 1,347     | 1,173     | 0      | 5,000       |
| Number of tags                  | 15        | 15        | 12        | 0      | 66          |
| Video age (days)                | 707       | 385       | 758       | 30     | 3,704       |
| Channel past engagement         | 0.038     | 0.036     | 0.015     | 0.005  | 0.130       |

Most of the count variables are highly skewed. The average video has about 1.3 million views, but the median is only about 230,000, which shows that a small number of very popular videos pull the mean up. The engagement rate is much more stable: half of all videos have an engagement rate between 2.5% and 4.9%, with a median of 3.5% (Figure 1). Because of the skew, subscribers, uploads and duration were log-transformed before modelling. Figure 2 shows how the 30% cut-off divides the dataset into the two classes.



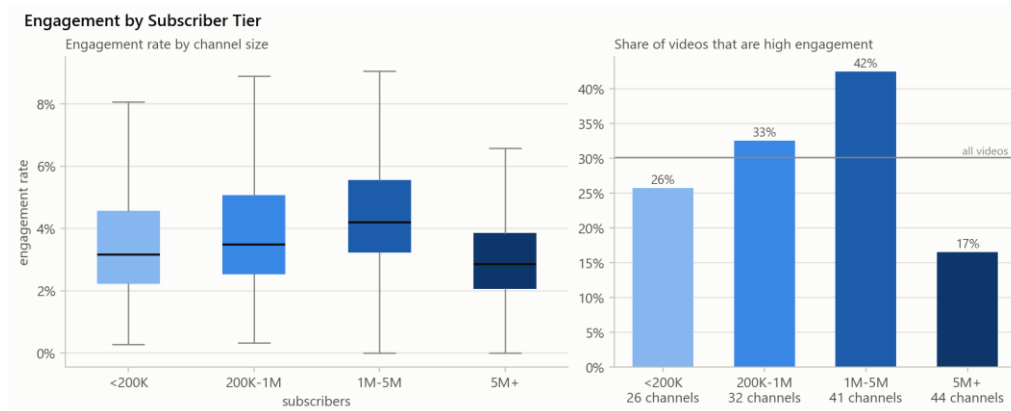
**Figure 1:** Distribution of engagement rate (log scale)



**Figure 2:** Class balance of the target variable in the cleaned dataset

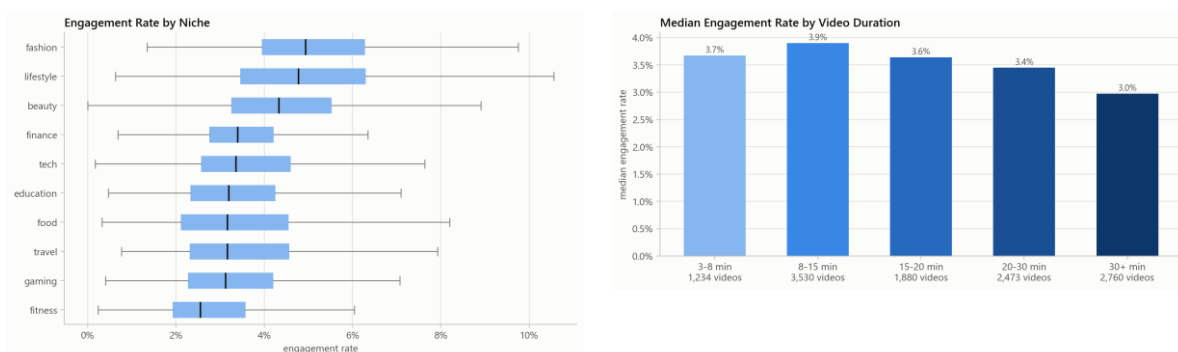
### 3.4 Exploratory Data Analysis

Figure 3 shows how engagement varies with channel size. The share of high-engagement videos increases from 26% for channels with fewer than 200,000 subscribers to 42% for channels with one to five million, but then drops to only 17% for channels above five million. So the largest channels, which usually charge the highest fees, have the lowest engagement. The Spearman correlation between subscriber count and engagement rate is -0.12, which is weak and slightly negative.



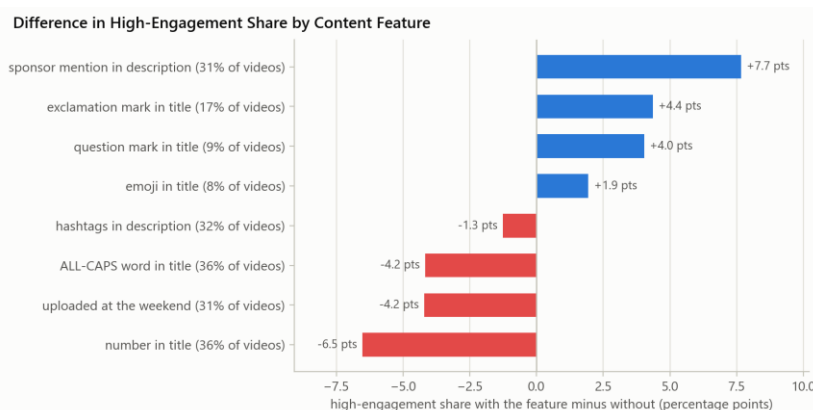
**Figure 3:** Engagement rate and share of high-engagement videos by subscriber tier

The niche of the channel makes a large difference (Figure 4, left). 60% of fashion videos and 53% of lifestyle videos are high engagement, while the figures for finance and fitness are only 19% and 13%. Video length also matters (Figure 4, right). Videos of 8 to 15 minutes have the highest median engagement rate (3.9%), and engagement goes down as videos get longer, with videos over 30 minutes having the lowest (3.0%).



**Figure 4:** Engagement rate by niche (left) and median engagement rate by video duration (right)

Figure 5 compares the share of high-engagement videos with and without each yes/no feature. Videos that mention a sponsor, discount code or affiliate link in the description are 7.7 percentage points more likely to be high engagement, which suggests that disclosed sponsorships do not reduce engagement. On the other hand, titles containing numbers (-6.5 points) or capitalised words (-4.2) and videos uploaded at the weekend (-4.2) are linked to lower engagement.



**Figure 5:** Difference in the share of high-engagement videos with and without each content feature

Upload time shows a clear pattern as well. Videos published between 12:00 and 23:59 UTC are high engagement 32% to 35% of the time, compared with 17% to 18% for videos published between 00:00 and 11:59 UTC, and Friday and Saturday are the weakest days. Title length, on the other hand, has

almost no relationship with engagement ( $\rho = -0.08$ ). It should be noted that these are simple comparisons that do not control for other factors. For example, well-established creators with loyal audiences are also more likely to get sponsorship deals.

Figure 6 shows the Spearman correlations between the numeric variables. I used Spearman correlation instead of Pearson because most of the variables are skewed. The channel's past engagement has by far the strongest relationship with engagement rate ( $\rho = 0.77$ ). All the other variables have weak correlations, the largest being video duration (-0.17) and the number of links in the description (0.16).

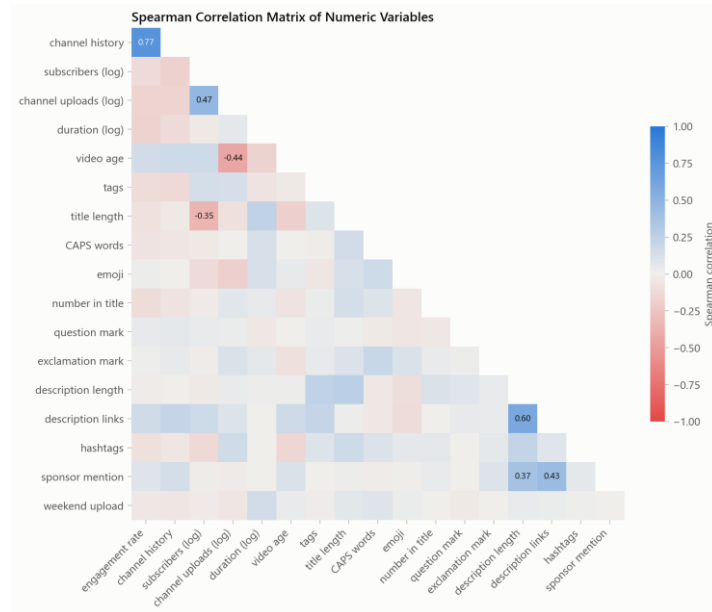


Figure 6: Spearman correlation matrix of the numeric variables

## 4. Analytics Method and Implementation

### 4.1 Choice of Method

The task is to predict whether a video will be high engagement or not, so it is a binary classification problem. I used two classification methods:

- **Logistic regression.** I chose this as the main interpretable model. Its coefficients show whether each factor increases or decreases the chance of high engagement and by how much, which makes it easy to turn the results into business recommendations.
- **Random forest.** A random forest combines the predictions of many decision trees. I used it because it can capture non-linear relationships and interactions between variables, such as the rise and then fall in engagement across subscriber tiers, which logistic regression cannot model directly.

I also built a simple logistic regression that uses subscriber count as its only input. This represents the way many brands choose influencers today, and gives a baseline that the other two models need to beat.

### 4.2 Implementation

The models were implemented in Python using pandas and scikit-learn [7] inside a Jupyter notebook. For logistic regression, the numeric variables were standardised, and the categorical variables (niche, upload day and time of day) were converted into dummy variables, with one category dropped as the reference level. The random forest used the numeric variables without scaling. Since only 30% of the videos are high engagement, both models were trained with balanced class weights so that they would not simply predict the majority class.

The data was split into 9,501 training videos (80%) and 2,376 test videos (20%) using a stratified split with `random_state = 42`, so that both parts kept the same proportion of high-engagement videos. The main hyperparameters, the regularisation strength `C` for logistic regression and `min_samples_leaf` for the random forest, were tuned with 5-fold cross-validation on the training data only. The best values were `C = 0.1` and `min_samples_leaf = 3` (Table 5). The test set was used only once, for the final evaluation.

**Table 5:** Hyperparameter tuning with 5-fold cross-validation on the training set

| Model               | Hyperparameter                  | Values tried            | Selected | CV ROC-AUC |
|---------------------|---------------------------------|-------------------------|----------|------------|
| Logistic regression | <code>C</code> (regularisation) | [0.01, 0.1, 1, 10, 100] | 0.1      | 0.897      |
| Random forest       | <code>min_samples_leaf</code>   | [1, 3, 10, 30]          | 3        | 0.921      |

### 4.3 Evaluation Approach

Because the classes are imbalanced, accuracy on its own can be misleading: a model that always predicts 'normal' would already be about 70% accurate (Figure 2). I therefore also report precision, recall, F1-score and ROC-AUC, together with the confusion matrices. In addition, I tested every model a second time using cross-validation in which all videos from the same channel are kept in the same fold. This checks whether the model works for creators it has never seen, which is the realistic situation when a brand is considering a new influencer.

## 5. Comparison with State-of-the-Art Methods

To place this work in context, I reviewed four recent published studies: three on predicting the popularity of online videos (two of them using YouTube data) and one on predicting the success of influencer marketing campaigns. Table 6 compares them with this case study.

**Table 6:** Comparison with published studies

| Published Study / Year     | Dataset                                                                                | Method Used                                                                         | Evaluation Metric                            | Key Result                                                                                                           | Comparison with This Study                                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nisa et al. (2021) [1]     | YouTube trending videos; 5,000 records with 5 selected features (from 15)              | XGBoost with feature selection and parameter tuning                                 | Accuracy, precision                          | Accuracy 86%, precision 64%; tuning improved both                                                                    | Also uses YouTube data and a tree-based model, but uses views, likes, dislikes and comments as inputs. My study excludes these because the target is calculated from them.                                                  |
| Ziyada & Shamoï (2024) [2] | Self-collected short environmental videos with daily likes, views, shares and comments | Ridge regression classifier with sentiment, emotion and basic video features        | Accuracy                                     | 0.80 using viewer comments; 0.67 using video features only                                                           | Similar way of creating classes (median split there, top 30% here). Their best features are viewer comments, which exist only after posting; my model uses pre-investment data only.                                        |
| Sourya et al. (2025) [3]   | Instagram influencer profiles (location, cost, reach, engagement, hashtags)            | Machine learning to recommend influencers for a given budget, location and category | Not stated in the publicly available summary | Recommends influencers that fit a campaign brief                                                                     | Same business goal of choosing influencers before spending, but on Instagram and at account level. My study predicts engagement for individual YouTube videos and tests on unseen creators.                                 |
| Chang & Chien (2025) [4]   | 52,867 videos from 370 YouTube channels published in 2024 (34,129 after cleaning)      | AdaBoost, decision tree, KNN, MLP, QDA, random forest and XGBoost                   | Accuracy                                     | XGBoost best with 99.24% accuracy; top predictors were average daily views, comment count and time since publication | Closest to mine in data source and size, and also compares several classifiers including random forest. Its top predictors are recorded after publication, so it solves a different task from my pre-investment prediction. |

The scores in these studies cannot be compared directly with mine, because each study uses a different dataset, target and experimental setup, so a higher reported accuracy does not mean a better method. The more useful comparison is in how the problem is approached. The three video popularity studies [1, 2, 4] predict popularity based on views, and their most useful inputs are response data such as likes, comments, daily views or viewer comments, which only exist after a video has been published. This is also why very high accuracies, such as the 99.24% reported in [4], are possible. My study instead predicts the engagement rate, which can be compared across channels of different sizes, and uses only information that is available before a brand invests. This makes the task harder, but it matches the decision a marketer actually has to make.

The study by Sourya et al. [3] is closest to mine in its business goal of helping brands choose influencers, but it works with Instagram profiles and recommends accounts rather than predicting engagement for individual videos. A recent review of 69 computational influencer marketing studies [5] also points out that research in this area often lacks explainability and well-documented datasets. I tried to address both points by using an interpretable logistic regression alongside the random forest and by documenting the data collection in full. Finally, the published descriptions of these studies do not mention testing on creators left out of training, which I included in this study.

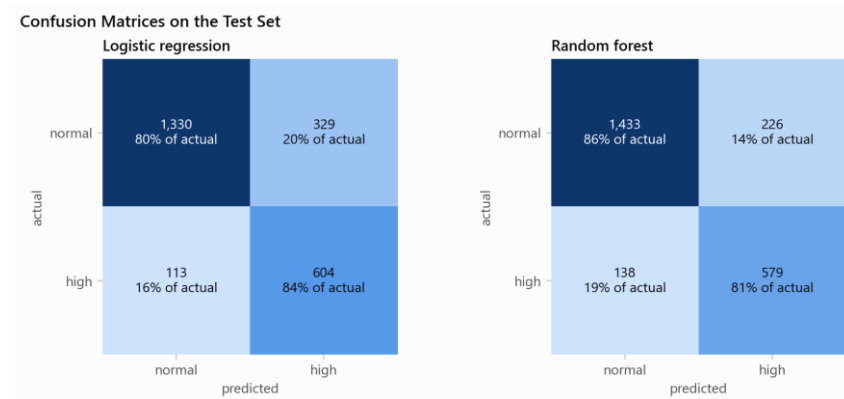
## 6. Results, Business Insights and Recommendations

### 6.1 Model Results

**Table 7:** Performance on the test set

| Model                 | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|-----------------------|----------|-----------|--------|----------|---------|
| Subscriber count only | 0.559    | 0.332     | 0.453  | 0.383    | 0.569   |
| Logistic regression   | 0.814    | 0.647     | 0.842  | 0.732    | 0.897   |
| Random forest         | 0.847    | 0.719     | 0.808  | 0.761    | 0.919   |

Table 7 shows the results on the test set. The random forest gave the best overall performance, with an ROC-AUC of 0.92 and an F1-score of 0.76. Its precision of 72% means that when it predicts a video will be high engagement, it is correct 72% of the time, which is about 2.4 times the base rate of 30%. It also identified 81% of all high-engagement videos. Logistic regression was slightly behind (ROC-AUC 0.90) but had a higher recall of 84%, at the cost of more false positives. The model based only on subscriber count had an ROC-AUC of 0.57, only a little better than random guessing (0.50), and a precision of 33%, almost the same as the base rate. This confirms that subscriber count on its own is not a useful predictor of engagement.

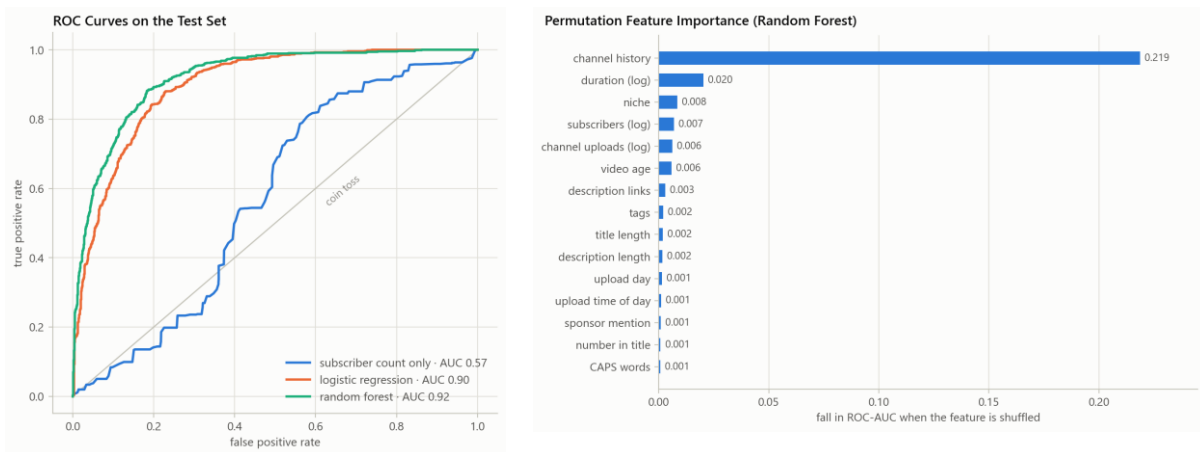


**Figure 7:** Confusion matrices of the two models on the test set

**Table 8:** ROC-AUC with standard 5-fold CV and with grouped 5-fold CV on unseen channels

| Model                             | Standard 5-fold CV | Unseen channels |
|-----------------------------------|--------------------|-----------------|
| Subscriber count only             | 0.562              | 0.554           |
| Logistic regression               | 0.898              | 0.885           |
| Random forest                     | 0.923              | 0.879           |
| Random forest, no channel history | 0.922              | 0.635           |

Table 8 compares the ROC-AUC under standard cross-validation with the grouped version, where the test channels are never seen in training. Both full models keep most of their performance on new creators (logistic regression 0.885, random forest 0.879). An interesting result came from a random forest trained without the channel history variable. It scored 0.922 in standard cross-validation, almost the same as the full model, but only 0.635 on unseen channels. Without the history variable, the model was mostly recognising channels from their exact subscriber and upload counts, and was not learning patterns that carry over to new creators.

**Figure 8:** ROC curves on the test set (left) and permutation importance for the random forest (right)

The permutation importance in Figure 8 confirms that channel history is the most important variable. Shuffling it reduces ROC-AUC by 0.22, about 11 times more than the next most important variable, video duration. In the logistic regression model, an increase of one standard deviation in channel history multiplies the odds of high engagement by 9.5. Longer videos reduce the odds (odds ratio 0.62), uploading between 12:00 and 17:59 UTC increases them (odds ratio 1.64 compared with 00:00 to 05:59), and subscriber count has almost no effect (odds ratio 1.04).

## 6.2 Key Business Insights

1. **Subscriber count is a poor basis for selecting influencers.** On its own it predicts engagement only slightly better than chance, and the largest channels (above five million subscribers) have the lowest share of high-engagement videos (17%).
2. **A creator's past engagement is the best predictor of future engagement.** It is by far the most important variable in both models.
3. **Mid-sized creators perform best.** Channels with one to five million subscribers have the highest share of high-engagement videos (42%).
4. **Engagement depends strongly on the niche,** ranging from 60% high-engagement videos in fashion to 13% in fitness.
5. **Video format and timing have a smaller but useful effect.** Videos of 8 to 15 minutes, and videos published on weekdays between 12:00 and 23:59 UTC, tend to perform better.
6. **Sponsorship mentions are not linked to lower engagement,** while title features such as length, emoji and capital letters add very little to the prediction.

## 6.3 Recommendations

1. Shortlist influencers by their recent engagement rate instead of their subscriber count. Before signing a creator, the brand can calculate the average (likes + comments) / views of their last five or more long-form videos from public data, and give priority to creators above about 4.5%.
2. Use mid-sized creators (one to five million subscribers) for campaigns where engagement is the goal. Use the largest creators mainly for awareness campaigns, and negotiate their price on cost per engagement rather than subscriber count.
3. Set engagement targets separately for each niche instead of using one benchmark for all campaigns.
4. Ask creators to place sponsored segments in videos of about 8 to 15 minutes, and schedule the release between Monday and Thursday, 12:00 to 23:59 UTC, avoiding Friday and Saturday.
5. Use the random forest model as a screening tool. On the test data, 72% of the videos it selected were high engagement, compared with 30% for random selection and 33% for selection by subscriber count. For the same budget, this would give roughly 2.4 times as many high-engagement placements as random selection.

## 7. Conclusion and References

### 7.1 Conclusion

In this case study I collected 22,807 YouTube videos from 150 public channels using the YouTube Data API, and analysed 11,877 long-form videos from 143 channels after cleaning. The main finding is that subscriber count, which is the measure most brands rely on, says very little about whether a video will engage its audience. The most useful information is the creator's own engagement history, followed by the niche, the length of the video and the time it is published. Using only information available before investing, the random forest model reached an ROC-AUC of 0.92 on the test set and 0.88 on creators it had not seen, and 72% of its positive predictions were correct against a base rate of 30%. These results support basing influencer selection on engagement history rather than audience size.

### 7.2 Limitations and Future Work

The study has some limitations. The channels were chosen by hand or through search rather than at random, so the results apply to these 143 English-language channels and not necessarily to YouTube as a whole. The relationships found in the exploratory analysis are associations and do not prove cause and effect. Engagement is also only a proxy for campaign success, since the data does not include clicks, sales or campaign costs. Finally, the statistics are a snapshot taken at the time of collection. Future work could use a random sample of channels, build a separate model for YouTube Shorts, and link engagement to sales data from real campaigns.

### 7.3 References

- [1] Nisa, M. U., Mahmood, D., Ahmed, G., Khan, S., Mohammed, M. A., & Damaševičius, R. (2021). Optimizing prediction of YouTube video popularity using XGBoost. *Electronics*, 10(23), 2962. <https://doi.org/10.3390/electronics10232962>
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